

9C The polar form of a complex number

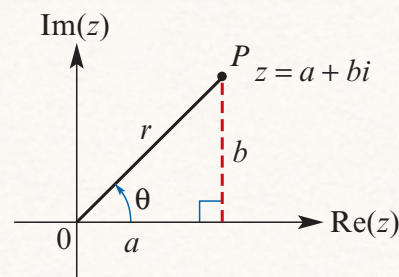
In the preceding sections, we have expressed complex numbers in Cartesian form. Another way of expressing complex numbers is using polar form.

Each complex number may be described by an angle and a distance from the origin. In this section, we will see that this is a very useful way to describe complex numbers.

► Polar form

The diagram shows the point P corresponding to the complex number $z = a + bi$. We see that $a = r \cos \theta$ and $b = r \sin \theta$, and so we can write

$$\begin{aligned} z &= a + bi && \leftarrow \text{Cartesian form} \\ &= r \cos \theta + (r \sin \theta) i \\ &= r(\cos \theta + i \sin \theta) \end{aligned}$$



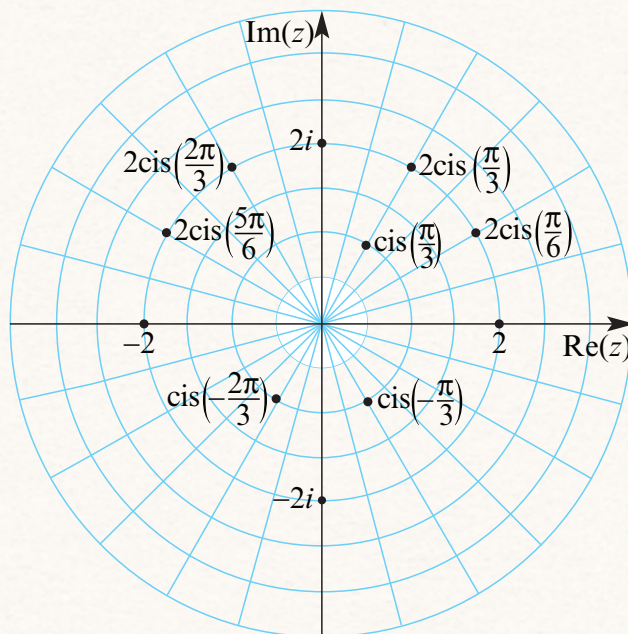
This is called the **polar form** of the complex number. The polar form is abbreviated to

$$z = r \operatorname{cis} \theta$$

- The distance $r = \sqrt{a^2 + b^2}$ is called the **modulus** of z and is denoted by $|z|$.
- The angle θ , measured anticlockwise from the horizontal axis, is called the **argument** of z and is denoted by $\arg z$.

Polar form for complex numbers is also called **modulus–argument form**.

This Argand diagram uses a polar grid with rays at intervals of $\frac{\pi}{12} = 15^\circ$.



► Non-uniqueness of polar form

Each complex number has more than one representation in polar form.

Since $\cos \theta = \cos(\theta + 2n\pi)$ and $\sin \theta = \sin(\theta + 2n\pi)$, for all $n \in \mathbb{Z}$, we can write

$$z = r \operatorname{cis} \theta = r \operatorname{cis}(\theta + 2n\pi) \quad \text{for all } n \in \mathbb{Z}$$

The convention is to use the angle θ such that $-\pi < \theta \leq \pi$.

Principal value of the argument

For a non-zero complex number z , the argument of z that belongs to the interval $(-\pi, \pi]$ is called the **principal value** of the argument of z and is denoted by $\text{Arg } z$. That is,

$$-\pi < \text{Arg } z \leq \pi$$

Example 10**Arg(z)**

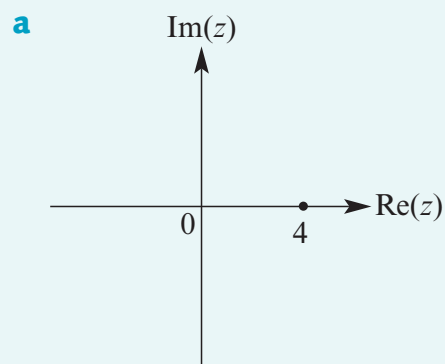
Find the modulus and principal argument of each of the following complex numbers:

a 4

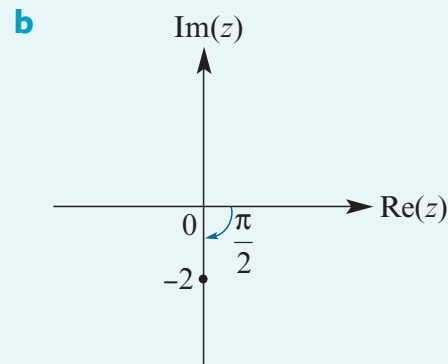
b $-2i$

c $1 + i$

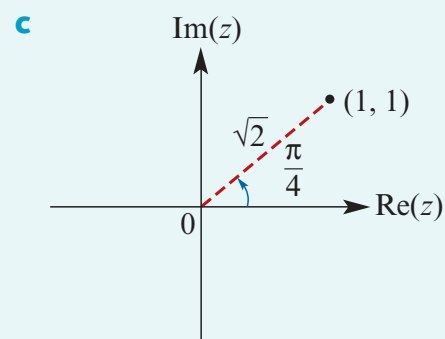
d $4 - 3i$

Solution

$$|4| = 4, \quad \text{Arg}(4) = 0$$

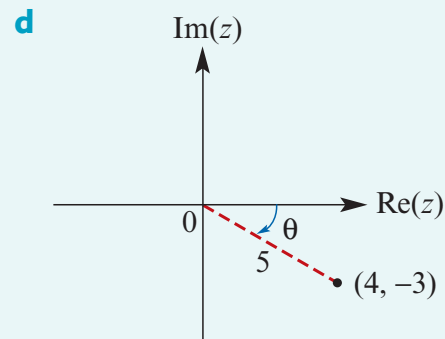


$$|-2i| = 2, \quad \text{Arg}(-2i) = -\frac{\pi}{2}$$



$$\begin{aligned} |1 + i| &= \sqrt{1^2 + 1^2} \\ &= \sqrt{2} \end{aligned}$$

$$\text{Arg}(1 + i) = \frac{\pi}{4}$$



$$\begin{aligned} |4 - 3i| &= \sqrt{4^2 + (-3)^2} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \text{Arg}(4 - 3i) &= -\tan^{-1}\left(\frac{3}{4}\right) \\ &\approx -0.64 \text{ radians} \end{aligned}$$



Using the TI-Nspire CX non-CAS

- To find the modulus of a complex number, use **menu** > **Number** > **Complex Number Tools** > **Magnitude**.

Alternatively, use $|\square|$ from the 2D-template palette $\left[\frac{\square}{\square} \right]$ or type `abs(`.

- To find the principal value of the argument, use **menu** > **Number** > **Complex Number Tools** > **Polar Angle**.

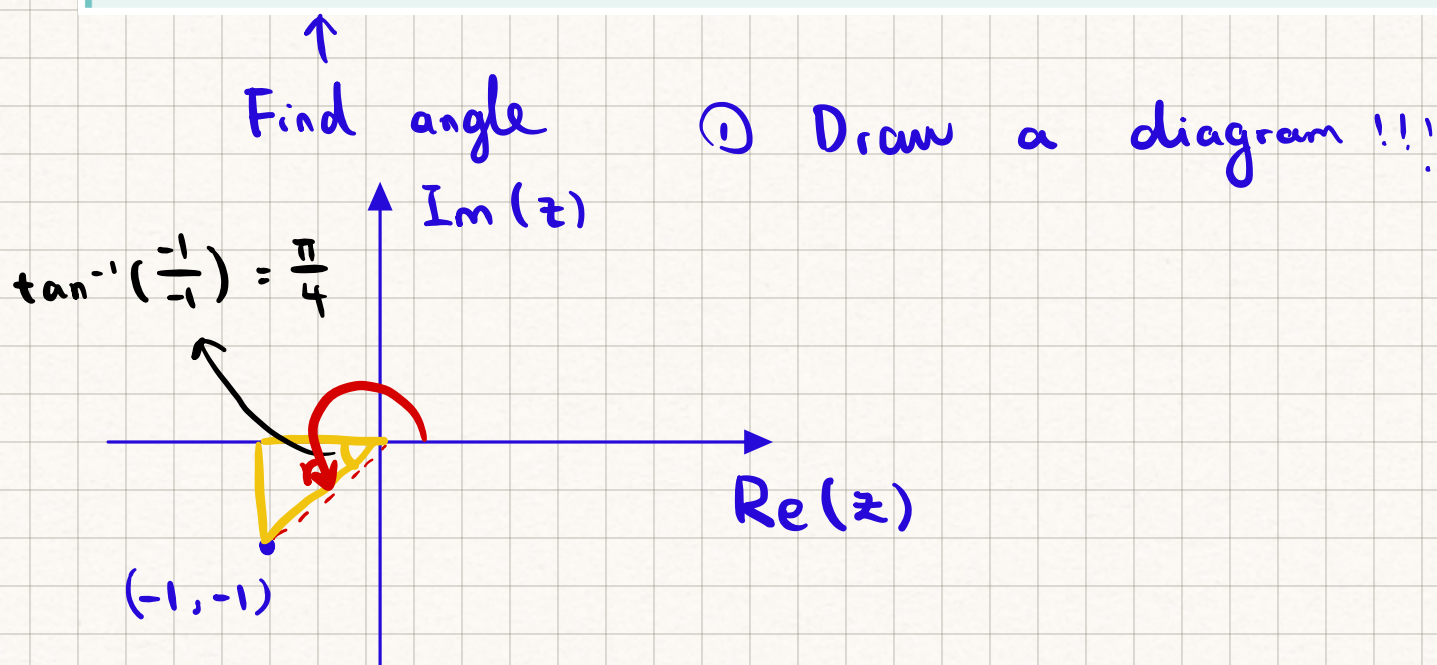
Note: Use π to access i .

1.1	SM3&4.....	RAD
angle(4)	0	
$ -2 \cdot i $	2	
angle(-2 · i)	-1.5708	
$ 1+i $	1.41421	
angle(1+i)	0.785398	

1.1	1.2	SM3&4.....	RAD
angle($4-3 \cdot i$)		-0.643501	
$ 4-3 \cdot i $		5	

Example 11

Find the argument of $-1 - i$ in the interval $[0, 2\pi]$.



$$\text{let } z = -1 - i$$

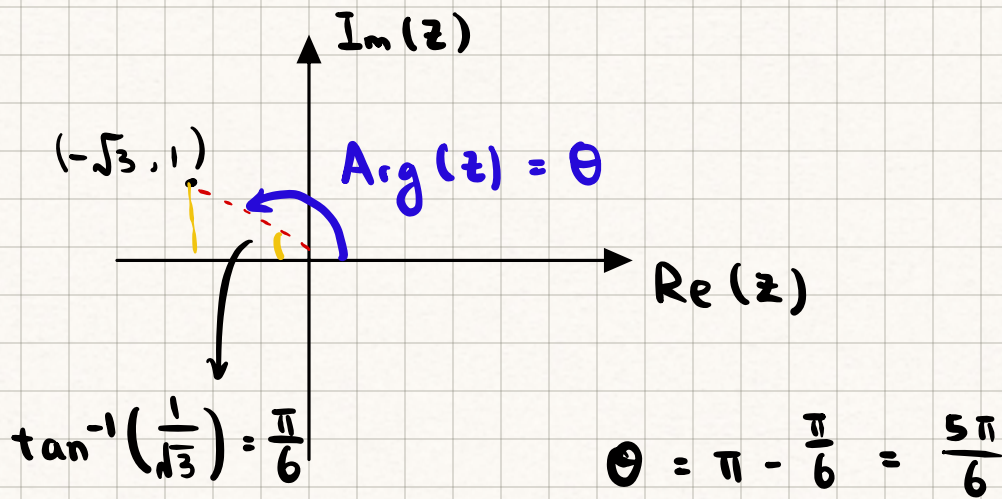
$$r = |z| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\arg(z) = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$$

Example 12

Express $-\sqrt{3} + i$ in the form $r \operatorname{cis} \theta$, where $\theta = \operatorname{Arg}(-\sqrt{3} + i)$.

$$r = \sqrt{(-\sqrt{3})^2 + 1^2} = 2$$



$$\Rightarrow -\sqrt{3} + i = 2 \operatorname{cis}\left(\frac{5\pi}{6}\right)$$

Example 13

Express $2 \operatorname{cis}\left(-\frac{3\pi}{4}\right)$ in the form $a + bi$.

$$r = 2 \quad \theta = -\frac{3\pi}{4}$$

$$a = r \cos \theta = 2 \cos\left(-\frac{3\pi}{4}\right) = -\sqrt{2}$$

$$b = r \sin \theta = 2 \sin\left(-\frac{3\pi}{4}\right) = -\sqrt{2}$$

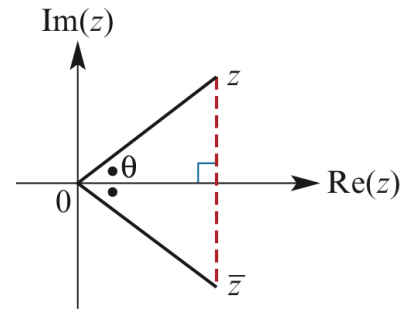
$$2 \operatorname{cis}\left(-\frac{3\pi}{4}\right) = -\sqrt{2} - \sqrt{2}i$$

► Complex conjugate in polar form

It is easy to show that the complex conjugate, $\bar{z}$, is a reflection of the point z in the horizontal axis.

Therefore, if $z = r \operatorname{cis} \theta$, then $\bar{z} = r \operatorname{cis}(-\theta)$.

In cartesian : $z = a + bi$
 $\bar{z} = a - bi$



Exercise 9C

HW

5F

Example 10

1 Find the modulus and principal argument of each of the following complex numbers:

a -3

b $5i$

c $i - 1$

d $\sqrt{3} + i$

e $2 - 2\sqrt{3}i$

f $(2 - 2\sqrt{3}i)^2$

2 Find the principal argument of each of the following, correct to two decimal places:

a $5 + 12i$

b $-8 + 15i$

c $-4 - 3i$

d $1 - \sqrt{2}i$

e $\sqrt{2} + \sqrt{3}i$

f $-(3 + 7i)$

Example 11

3 Find the argument of each of the following in the interval stated:

a $1 - \sqrt{3}i$ in $[0, 2\pi]$

b $-7i$ in $[0, 2\pi]$

c $-3 + \sqrt{3}i$ in $[0, 2\pi]$

d $\sqrt{2} + \sqrt{2}i$ in $[0, 2\pi]$

e $\sqrt{3} + i$ in $[-2\pi, 0]$

f $2i$ in $[-2\pi, 0]$

4 Convert each of the following arguments into principal arguments:

a $\frac{5\pi}{4}$

b $\frac{17\pi}{6}$

c $\frac{-15\pi}{8}$

d $\frac{-5\pi}{2}$

Example 12

5 Convert each of the following complex numbers from Cartesian form $a + bi$ into the form $r \operatorname{cis} \theta$, where $\theta = \operatorname{Arg}(a + bi)$:

a $-1 - i$

b $\frac{1}{2} - \frac{\sqrt{3}}{2}i$

c $\sqrt{3} - \sqrt{3}i$

d $\frac{1}{\sqrt{3}} + \frac{1}{3}i$

e $\sqrt{6} - \sqrt{2}i$

f $-2\sqrt{3} + 2i$

Example 13

6 Convert each of the following complex numbers into the form $a + bi$:

a $2 \operatorname{cis}\left(\frac{3\pi}{4}\right)$

b $5 \operatorname{cis}\left(\frac{-\pi}{3}\right)$

c $2\sqrt{2} \operatorname{cis}\left(\frac{\pi}{4}\right)$

d $3 \operatorname{cis}\left(\frac{-5\pi}{6}\right)$

e $6 \operatorname{cis}\left(\frac{\pi}{2}\right)$

f $4 \operatorname{cis} \pi$

7 Let $z = \operatorname{cis} \theta$. Show that:

a $|z| = 1$

b $\frac{1}{z} = \operatorname{cis}(-\theta)$

8 Find the complex conjugate of each of the following:

a $2 \operatorname{cis}\left(\frac{3\pi}{4}\right)$

b $7 \operatorname{cis}\left(\frac{-2\pi}{3}\right)$

c $-3 \operatorname{cis}\left(\frac{2\pi}{3}\right)$

d $5 \operatorname{cis}\left(\frac{-\pi}{4}\right)$

► De Moivre's theorem

De Moivre's theorem allows us to readily simplify expressions of the form z^n when z is expressed in polar form.

De Moivre's theorem

$$(r \operatorname{cis} \theta)^n = r^n \operatorname{cis}(n\theta), \text{ where } n \in \mathbb{Z}$$

9G Roots of complex numbers

Equations of the form $z^n = a$, where $a \in \mathbb{C}$, are often solved by using De Moivre's theorem.

Write both z and a in polar form, as $z = r \operatorname{cis} \theta$ and $a = q \operatorname{cis} \varphi$.

Then $z^n = a$ becomes

$$(r \operatorname{cis} \theta)^n = q \operatorname{cis} \varphi$$

$$\therefore r^n \operatorname{cis}(n\theta) = q \operatorname{cis} \varphi \quad (\text{using De Moivre's theorem})$$

Compare modulus and argument:

$$r^n = q \quad \operatorname{cis}(n\theta) = \operatorname{cis} \varphi$$

$$r = \sqrt[n]{q} \quad n\theta = \varphi + 2k\pi \quad \text{where } k \in \mathbb{Z}$$

$$\theta = \frac{1}{n}(\varphi + 2k\pi) \quad \text{where } k \in \mathbb{Z}$$

This will provide all the solutions of the equation.



Example 27

Solve $z^3 = 1$.

$$\text{Let } z = r \operatorname{cis} \theta$$

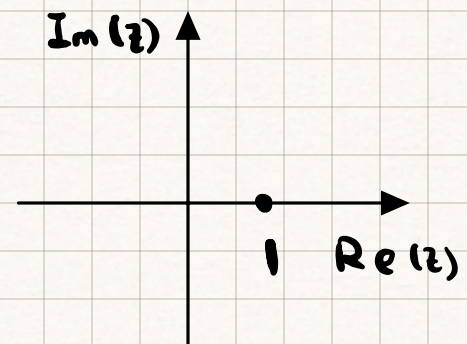
$$1 = 1 + 0i \quad \xrightarrow{\text{Polar form}} \quad = 1 \operatorname{cis} 0$$

$$z^3 = 1 \Rightarrow (r \operatorname{cis} \theta)^3 = 1 \operatorname{cis} 0$$

$$r^3 \operatorname{cis}(3\theta) = 1 \operatorname{cis} 0$$

$$\Rightarrow r^3 = 1 \quad \text{and} \quad 3\theta = 0 + 2k\pi, \quad k \in \mathbb{Z}$$

$$\Rightarrow r = 1 \quad \text{and} \quad \theta = \frac{2k\pi}{3}, \quad k \in \mathbb{Z}$$



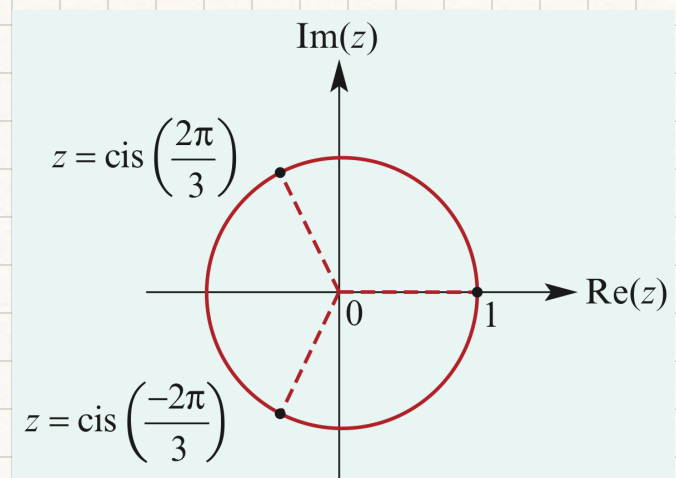
Start finding solutions.

- For $k = 0$, $r = 1$ and $\theta = 0$
 $z = 1 \text{ cis } 0$
- For $k = 1$, $r = 1$ and $\theta = \frac{2\pi}{3}$
 $z = 1 \text{ cis } \left(\frac{2\pi}{3}\right)$
- For $k = 2$, $r = 1$ and $\theta = \frac{4\pi}{3}$
 $z = 1 \text{ cis } \left(\frac{4\pi}{3}\right) = 1 \text{ cis } \left(-\frac{2\pi}{3}\right)$
- For $k = 3$, $r = 1$ and $\theta = 2\pi$
 $z = 1 \text{ cis } (2\pi) = 1 \text{ cis } (0)$

The solution begin to repeat

$$z = 1 \text{ cis } 0 , 1 \text{ cis } \left(\frac{2\pi}{3}\right) , 1 \text{ cis } \left(-\frac{2\pi}{3}\right)$$

Draw circle
radius r



Example 28

Solve $z^2 = 1 + i$.

Let $z = r \operatorname{cis} \theta$. Note that $1 + i = \sqrt{2} \operatorname{cis}(\frac{\pi}{4})$

$$(r \operatorname{cis} \theta)^2 = \sqrt{2} \operatorname{cis}(\frac{\pi}{4})$$

$$r^2 \operatorname{cis} 2\theta = \sqrt{2} \operatorname{cis}(\frac{\pi}{4})$$

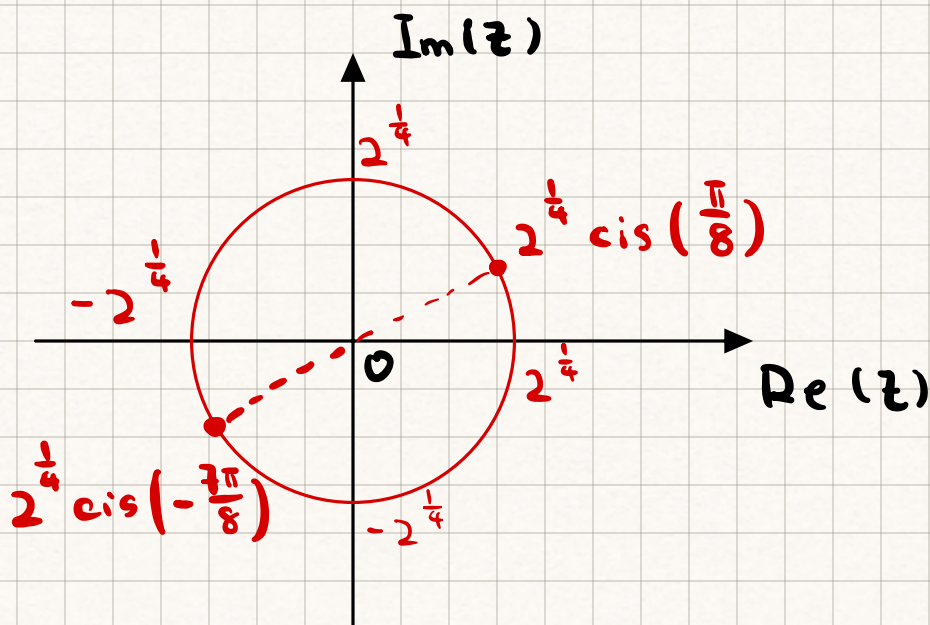
$$\Rightarrow r^2 = \sqrt{2} \quad \text{and} \quad 2\theta = \frac{\pi}{4} + 2k\pi, \quad k \in \mathbb{Z}$$

$$\Rightarrow r = 2^{\frac{1}{4}} \quad \text{and} \quad \theta = \frac{\pi}{8} + k\pi, \quad k \in \mathbb{Z}$$

Hence, $z = 2^{\frac{1}{4}} \operatorname{cis}(\frac{\pi}{8} + k\pi), \quad k \in \mathbb{Z}$.

* For $k = 0$, $z = 2^{\frac{1}{4}} \operatorname{cis}(\frac{\pi}{8})$

* For $k = 1$, $z = 2^{\frac{1}{4}} \operatorname{cis}(\frac{9\pi}{8}) = 2^{\frac{1}{4}} \operatorname{cis}(\frac{9\pi}{8} - 2\pi)$
 $= 2^{\frac{1}{4}} \operatorname{cis}(-\frac{7\pi}{8})$



Solutions of $z^n = a$

For $n \in \mathbb{N}$ and $a \in \mathbb{C}$, the solutions of the equation $z^n = a$ are called the **n th roots** of a .

- The solutions of $z^n = a$ lie on a circle with centre the origin and radius $|a|^{\frac{1}{n}}$.
- There are n solutions and they are equally spaced around the circle at intervals of $\frac{2\pi}{n}$.
This observation can be used to find all solutions if one is known.

1 For each of the following, solve the equation over $\mathbb{C}$ and show the solutions on an Argand diagram:

a $z^2 + 1 = 0$

b $z^3 = 27i$

c $z^2 = 1 + \sqrt{3}i$

d $z^2 = 1 - \sqrt{3}i$

e $z^3 = i$

f $z^3 + i = 0$

2 Find all the cube roots of the following complex numbers:

a $4\sqrt{2} - 4\sqrt{2}i$

b $-4\sqrt{2} + 4\sqrt{2}i$

c $-4\sqrt{3} - 4i$

d $4\sqrt{3} - 4i$

e $-125i$

f $-1 + i$